

Fiscal Foresight and Non-Invertibility

Leeper–Walker–Yang (2013), Section 3: Full Derivation

Companion notes for Unit 8

These notes derive the structural MA representation in Leeper, Walker, and Yang (2013, *Econometrica*) from first principles. The slides give the storyline; here we work out every step. The setup is the “analytical example” of their Section 3 (log utility, full depreciation, inelastic labor) — the simplest specification in which fiscal foresight produces a non-invertible MA.

1 Model

Preferences. A representative agent maximizes

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \log C_t, \quad \beta \in (0, 1).$$

Production. Output is Cobb–Douglas with full depreciation ($\delta = 1$) and inelastic labor absorbed into TFP:

$$Y_t = A_t K_{t-1}^{\alpha}, \quad \alpha \in (0, 1).$$

Resource constraint. Output is split between consumption, investment, and lump-sum government transfers $T_t = \tau_t Y_t$ (taxes are rebated):

$$C_t + K_t = (1 - \tau_t) Y_t + T_t = Y_t = A_t K_{t-1}^{\alpha}.$$

With full depreciation K_t is both this period’s investment and next period’s productive capital.

Shocks. TFP and tax shocks are mutually independent i.i.d. white noise:

$$a_t \equiv \log A_t - \log \bar{A} = \varepsilon_{A,t}, \quad \mathbb{E}[\varepsilon_{A,t}^2] = \sigma_A^2,$$

$$\hat{\tau}_t \equiv \tau_t - \tau = \varepsilon_{\tau,t-q}.$$

The second line is the foresight specification: a tax shock realized at time $t - q$ shows up as the tax-rate deviation $\hat{\tau}_t$ at time t . Equivalently, the news $\varepsilon_{\tau,t}$ that arrives at time t announces the tax rate at time $t + q$.

Information. The agent’s information set at time t is

$$\Omega_t = \{\varepsilon_{A,s}, \varepsilon_{\tau,s}\}_{s \leq t}.$$

In particular, at time t the agent knows the next q tax rates exactly: $\tau_{t+1}, \tau_{t+2}, \dots, \tau_{t+q}$ are all functions of past tax shocks $\varepsilon_{\tau,s}$ with $s \leq t$.

2 Agent's problem and Euler equation

The agent chooses $\{C_t, K_t\}$ to maximize expected log utility subject to the resource constraint. Substituting $C_t = A_t K_{t-1}^\alpha - K_t$ into the objective and taking first-order conditions with respect to K_t :

$$-\frac{1}{C_t} + \beta \mathbb{E}_t \left[\frac{\alpha A_{t+1} K_t^{\alpha-1} (1 - \tau_{t+1})}{C_{t+1}} \right] = 0.$$

(The factor $(1 - \tau_{t+1})$ arises because the agent's effective return on capital is the after-tax marginal product $\alpha(1 - \tau_{t+1})Y_{t+1}/K_t$; the tax revenue is rebated as T_{t+1} which enters the budget constraint but is taken as given.)

Rearranging:

$$\frac{1}{C_t} = \alpha \beta \mathbb{E}_t \left[\frac{(1 - \tau_{t+1}) Y_{t+1}}{K_t C_{t+1}} \right]. \quad (1)$$

Where anticipation enters. The conditional expectation $\mathbb{E}_t[\cdot]$ is taken with respect to Ω_t . When $q \geq 1$, the next-period tax rate τ_{t+1} is *known* at time t (it equals $\tau + \varepsilon_{\tau, t-q+1}$, a deterministic function of past shocks in Ω_t). The tax news $\varepsilon_{\tau, t}$ that arrives at time t moves τ_{t+q+1} — which then enters $\mathbb{E}_t[\cdot]$ via continuation values — and hence shifts today's capital choice K_t . This is the “foresight channel.”

3 Steady state and log-linearization

Steady state. With $\varepsilon_A = 0$, $\hat{\tau} = 0$:

$$\begin{aligned} \bar{K}^{1-\alpha} &= \alpha \beta (1 - \tau) \bar{A}, \\ \bar{C} &= \bar{A} \bar{K}^\alpha - \bar{K} = (1 - \alpha \beta (1 - \tau)) \bar{A} \bar{K}^\alpha. \end{aligned}$$

Define the key reduced-form parameter

$$\boxed{\theta \equiv \alpha \beta (1 - \tau).}$$

Note $\theta \in (0, 1)$ since $\alpha, \beta, \tau \in (0, 1)$. The steady-state capital–output ratio is $\bar{K}/\bar{Y} = \theta$, and the steady-state consumption–output ratio is $1 - \theta$.

Linearizing the Euler equation. Let $\hat{C}_t = \log(C_t/\bar{C})$, $\hat{K}_t = \log(K_t/\bar{K})$, $\hat{Y}_t = \log(Y_t/\bar{Y}) = a_t + \alpha \hat{K}_{t-1}$. Taking logs of (1) and expanding around the steady state:

$$-\hat{C}_t = -\hat{K}_t - \frac{\tau}{1 - \tau} \mathbb{E}_t \hat{\tau}_{t+1} + \mathbb{E}_t \hat{Y}_{t+1} - \mathbb{E}_t \hat{C}_{t+1}.$$

The resource constraint linearizes to

$$(1 - \theta) \hat{C}_t + \theta \hat{K}_t = \hat{Y}_t = a_t + \alpha \hat{K}_{t-1}.$$

Eliminating $\hat{C}_t, \hat{C}_{t+1}$ via the resource constraint yields a second-order difference equation in $\hat{K}_t$:

$$\mathbb{E}_t \hat{K}_{t+1} - (\theta + \alpha) \hat{K}_t + \alpha \theta \hat{K}_{t-1} = \frac{1}{\theta} \mathbb{E}_t [a_{t+1} - \theta^{-1} a_t] + \frac{1 - \theta}{\theta} \cdot \frac{\tau}{1 - \tau} \mathbb{E}_t \hat{\tau}_{t+1}. \quad (2)$$

Define $\kappa \equiv (1 - \theta)\tau/(1 - \tau)$ for the coefficient on the tax-anticipation term.

4 Saddlepath solution

The characteristic polynomial of the homogeneous part of (2) is

$$\lambda^2 - (\theta + \alpha)\lambda + \alpha\theta = (\lambda - \alpha)(\lambda - \theta),$$

with roots α (stable, $|\alpha| < 1$) and θ (stable, $|\theta| < 1$). With two predetermined-or-saddlepath roots (eigenvalue $\theta < 1$ is the relevant “forward” root), the unique non-explosive solution iterates the unstable root forward:

$$\hat{K}_t = \alpha \hat{K}_{t-1} + a_t - \kappa \sum_{i=0}^{\infty} \theta^i \mathbb{E}_t \hat{\tau}_{t+i+1}. \quad (3)$$

TFP is i.i.d., so $\mathbb{E}_t[a_{t+s}] = 0$ for $s \geq 1$ and the TFP contribution is just the current a_t . The tax-anticipation sum captures the present discounted value of expected future tax-rate deviations, discounted at rate θ .

5 Substituting the tax rule with foresight ($q = 2$)

Take $q = 2$ for concreteness. The tax rule $\hat{\tau}_t = \varepsilon_{\tau,t-2}$ implies, at time t ,

$$\mathbb{E}_t \hat{\tau}_{t+1} = \varepsilon_{\tau,t-1}, \quad \mathbb{E}_t \hat{\tau}_{t+2} = \varepsilon_{\tau,t}, \quad \mathbb{E}_t \hat{\tau}_{t+i} = 0 \text{ for } i \geq 3.$$

Only the $i = 0$ and $i = 1$ terms of (3)’s sum contribute:

$$\sum_{i=0}^{\infty} \theta^i \mathbb{E}_t \hat{\tau}_{t+i+1} = \varepsilon_{\tau,t-1} + \theta \varepsilon_{\tau,t}.$$

Substituting into (3):

$$\hat{K}_t - \alpha \hat{K}_{t-1} = \varepsilon_{A,t} - \kappa(\varepsilon_{\tau,t-1} + \theta \varepsilon_{\tau,t}).$$

Equivalently, with the lag operator L :

$$(1 - \alpha L) \hat{K}_t = \varepsilon_{A,t} - \kappa(L + \theta) \varepsilon_{\tau,t}. \quad (4)$$

6 The structural MA $M(L)$

Stack the observables and shocks:

$$y_t = \begin{pmatrix} a_t \\ \hat{K}_t \end{pmatrix}, \quad \varepsilon_t = \begin{pmatrix} \varepsilon_{\tau,t} \\ \varepsilon_{A,t} \end{pmatrix}.$$

From $a_t = \varepsilon_{A,t}$ and (4):

$$y_t = M(L) \varepsilon_t, \quad M(L) = \begin{pmatrix} 0 & 1 \\ -\frac{\kappa(L + \theta)}{1 - \alpha L} & \frac{1}{1 - \alpha L} \end{pmatrix}.$$

The entries have direct economic content:

- $M_{11}(L) = 0$: TFP a_t doesn’t respond to tax shocks at all (TFP is exogenous to tax policy).

- $M_{12}(L) = 1$: TFP responds one-for-one to its own shock (i.i.d.).
- $M_{21}(L) = -\kappa(L + \theta)/(1 - \alpha L)$: capital responds to the tax-news shock contemporaneously (the θ term, from anticipation) and at lag 1 (the L term, from implementation when $q = 2$); the denominator $(1 - \alpha L)$ is the capital-accumulation AR factor.
- $M_{22}(L) = 1/(1 - \alpha L)$: capital responds to TFP through the AR dynamics.

7 Non-invertibility

Compute the determinant. The $(1, 1)$ entry is zero, so the determinant simplifies along the top row:

$$\det M(L) = 0 \cdot M_{22}(L) - 1 \cdot M_{21}(L) = \frac{\kappa(L + \theta)}{1 - \alpha L}.$$

Setting $\det M(z) = 0$ gives a zero at $z = -\theta$. The pole at $z = 1/\alpha$ lies outside the unit disk (since $\alpha < 1$), so the relevant root is

$$z^* = -\theta, \quad |z^*| = \theta \in (0, 1).$$

The MA root sits strictly inside the unit disk. Hence $M(L)$ is non-invertible, and by the bridge result on the slides ($C(L) = \Phi(L)M(L)$ with $\det \Phi$ no-inside-roots) so is $C(L)$. A VAR fit to $y_t = (a_t, \hat{K}_t)'$ cannot recover $(\varepsilon_{\tau,t}, \varepsilon_{A,t})$ from current and past y .

8 Economic interpretation

The MA root is $|z^*| = \theta = \alpha\beta(1 - \tau)$. What controls its position?

- Larger α (capital-intensive production), larger β (patient agents), or smaller τ (lower tax distortion) all push θ closer to 1 — the root nears the unit circle, and non-invertibility softens.
- For standard macro calibrations ($\alpha \approx 0.3$, $\beta \approx 0.99$, $\tau \approx 0.25$), $\theta \approx 0.22$ — the root sits well inside the unit disk, and the non-invertibility is severe.

The economics: θ measures how strongly capital (the forward-looking variable) responds to anticipated tax news. When FL response is strong, the news is well-reflected in current capital and the VAR can detect it; when FL response is weak relative to the implementation effect at $t + q$, the VAR cannot disentangle current news from past news. Within LWY's specification, the forward-looking response can never be strong enough — the economic restrictions $\alpha, \beta, \tau \in (0, 1)$ keep θ bounded away from 1, and non-invertibility is robust.

9 Connection to identification

The non-invertible MA implies that the Wold innovation u_t recovered by fitting a VAR to y_t is *not* a contemporaneous rotation of $(\varepsilon_{\tau,t}, \varepsilon_{A,t})$. Instead, u_t is a “Blaschke-flipped” linear combination involving current and past structural shocks. No Σ_u -based identification scheme (Cholesky, Blanchard–Quah, sign restrictions) can recover the news shock from u alone. An external news-timed instrument (proxy SVAR) recovers the relevant column of the structural impact matrix directly.

References

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